

Taxing Paradise: Optimal Commodity Taxation with Tourists

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Abstract

I develop a theory of the jointly optimal choice of nonlinear income taxes and commodity taxes which accounts for the presence of tourists. The Pareto efficient uniform commodity tax rate is the one that maximizes revenue from tourists; impacts of the tax on residents are irrelevant, as they can always be compensated through income tax reductions. In addition, the presence of tourists in the tax base overturns the classic Atkinson-Stiglitz result implying uniform commodity taxation is Pareto efficient. Efficient rate differentiation is characterized by a Corlett–Hague-style rule which requires placing lower taxes on goods which tourists can more easily substitute for expenditures outside of the destination economy. Notably, the impact of commodity taxes on residents does not influence the pattern of efficient differentiation, nor the overall level of commodity taxation: resident consumption behavior only influences the magnitude of efficient rate differentiation. The endogeneity of tourist arrivals may also provide a rationale for rate differentiation: goods should be taxed at a lower rate if they are disproportionately consumed by tourists who are relatively more likely to forgo visiting as prices rise. These results do not support the intuitive idea it is optimal to levy higher rates on goods that are disproportionately consumed by tourists. On the other hand, they do rationalize some commonly observed tax policies in tourist destinations, such as higher rates on hotel accommodation and car rentals.

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It is a common view in tourism-dependent economies that as much of the tax burden as possible should be shifted onto tourists. In the United States, this often manifests in tax policy that levies higher tax rates on goods that are more heavily consumed by tourists: twenty-five states levy taxes on hotel rooms and other visitor accommodations at higher rates than other goods [HVS, 2022].

On the other hand, in some jurisdictions, concerns about competition appear to have been used to justify levying lower rates on goods that tourists buy. According to a 2014 OECD report, hotel accommodations are one of the most common goods benefiting from reduced VAT rates: twenty-two OECD countries did so at the time [OECD, 2014].

Both these policy responses suggest that tourist demand provides a rationale for departing from the implications of the Atkinson-Stiglitz theorem [Atkinson and Stiglitz, 1976], which implies that as long as the government can also levy nonlinear income taxes, then, uniform commodity taxation is Pareto efficient (at least, under certain restrictions on preferences). Prior literature has proposed several justifications for departing from uniformity, including departures from weak separability between consumption and leisure [Christiansen, 1984], preference heterogeneity [Saez, 2002, Ferey et al., 2022], behavioral biases [Farhi and Gabaix, 2020], credit constraints [Boadway et al., 2019], income tax evasion [Boadway et al., 1994, Gerritsen, 2025], goods that also generate pre-tax income [Goldin et al., 2026], multidimensional heterogeneity in unobserved endowments [Cremer et al., 2001], heterogeneous returns to saving [Gerritsen et al., 2025], and non-tax filing [Moore, 2026]. This paper adds a rationale based on tourism demand.

In so doing, I clarify the interpretation of two prior papers [Hämäläinen, 2004, Gooroochurn, 2009] that explore optimal commodity taxation in the presence of tourists within a Ramsey framework (see also Bird 1992 for an earlier treatment of tourism taxation in developing economies). Both papers show that taxes should be higher on goods that are more heavily consumed by tourists. However, these results are derived in a model without nonlinear income taxes. My findings show that these results are driven by a restriction on the set of tax instruments available to the planner. As I shall show, when the planner can adjust a nonlinear income tax levied on residents, optimal tax rates are in fact not relatively high for goods where tourists account for a larger share of demand. The reason is simple: if residents can be compensated for the costs commodity taxes impose on them, then commodity taxes can be set solely for the purpose of maximally extracting revenue from tourists.

My results also connect to the literature on trade policy. Since it is the consumer, rather than the good, that crosses the border, tourist purchases can be viewed as exports of non-traded services, and the visitor-facing component of a commodity tax acts as an export tariff. What distinguishes the tourism setting from classical treatments of export pricing [Dixit and Norman, 1980], and from recent analyses of trade policy with domestic

redistribution [Costinot and Werning, 2023, Hosseini and Shourideh, 2024], is that this tariff is not a separate instrument: in the setting I study, the destination levies the same commodity tax on residents and visitors alike. A second difference is the central role of extensive margin responses—the decision to visit at all—which existing analyses of export pricing do not consider.

In section 1, I provide a simple benchmark result showing that the Pareto efficient uniform commodity tax rate is the one which maximizes the revenue obtained from tourists. From the perspective of residents, there exists a continuum of different tax systems, which are both welfare and revenue-equivalent for residents. Pareto efficient taxation thus calls for choosing from amongst these tax systems the one that levies tourist revenue maximizing commodity tax. Resident demand is thus rendered irrelevant to the determination of the optimal uniform tax rate.

In section 2, I turn to the case of non-uniform taxation. For simplicity, I focus on the case with two taxable goods and preferences that, absent tourist demand, would imply that optimal commodity taxes are uniform [Atkinson and Stiglitz, 1976]. The presence of tourist demand breaks this result: optimal rates vary across goods, following a Corlett-Hague-style rule. My results show that rates should be lower on goods which, for tourists, are more substitutable with spending outside the destination economy, which the destination cannot tax. This mirrors the findings of Ramsey [1927] and Corlett and Hague [1953] that, without nonlinear income taxes, commodity taxes should be lower on goods which are more substitutable with untaxed leisure. This result parallels those seen in other papers that break the Atkinson-Stiglitz theorem by introducing a population which is not subject or imperfectly subject to a nonlinear income tax [Boadway et al., 1994, 2019, Gerritsen, 2025, Moore, 2026].

An analysis of tourist taxation is incomplete without allowing for the possibility that taxes may induce tourists to skip a visit if they make a trip to the destination too costly. In section 2.3, I show that extensive margin responses can create a further rationale for rate differentiation when they vary across visitors. All else equal, taxes should be lower on those goods which are more heavily demanded by tourists who are more likely to abandon a visit as its cost rises: relative consumption patterns act as tags for extensive margin responsiveness.

In section 2.4, I show that the optimal average commodity tax rate paid by tourists is characterized by a kind of inverse elasticity rule. The elasticity in question is a revenue-weighted average of the rates at which the visitor tax bases leak out of the destination as all destination prices rise together, counting both substitution toward outside consumption by continuing visitors and the loss of visitors who do not come at all. Importantly, this average tax rate formula only includes parameters measuring tourist responses to commodity taxation. Thus, although resident demand patterns play a role in determining the magnitude of

optimal rate differentiation, they play no direct role in determining the optimal level of commodity taxation, generalizing my findings regarding the case of a uniform tax in section 1.

For simplicity, all these results are derived in a partial equilibrium model. Section 2.5 shows that they can be reinterpreted through the lens of a simple general equilibrium model of a small open economy with endogenous prices and wages.

Section 3 provides a qualitative discussion of the potential policy implications of these results, while also highlighting directions for future extensions of this work. In particular, I note the importance of accounting for externalities, and for the richer general equilibrium effects.

1 Optimal Uniform Commodity Tax

Consider an economy with a continuum of residents each indexed by some $i \in [0, 1]$. Each resident i chooses taxable income z_i to maximize their utility

$$u_i(c, z)$$

subject to the budget constraint

$$(1 + t)c = z - T(z)$$

where $T(\cdot)$ is a nonlinear income tax and t is a linear (uniform) commodity tax rate. Here, c is taken to represent the residents' total consumption across many goods. With a uniform commodity tax, this is without loss of generality.

Now suppose that commodity demand in this economy stems not only from residents, but also from tourists (or other non-residents). Let C_V be the aggregate spending of these *visitor* taxpayers on commodities in the destination economy. Further, let $C_R \equiv \int_0^1 c_i di$ be the aggregate spending of resident taxpayers on commodities.¹ Total tax revenue in this economy is then

$$R \equiv \int_0^1 T(z_i) di + t(C_R + C_V).$$

Here and throughout, tourists carry no weight in the planner's objective: a tax system is "Pareto efficient" if no reform can make some residents better off without making others worse off.² Implicit in this definition is

¹For simplicity, I have assumed that residents do not themselves spend some of their income as tourists in other destinations. Relaxing this assumption would imply a somewhat lower optimal rate.

²The results would need only minor modification if tourists had positive welfare weights: optimal rates would be lower. What drives the analysis is not the zero weight but the fact that tourists cannot be subjected to the destination's income tax. The extreme assumption is also natural under a political economy reading: "tourists don't vote" [Mak, 2003].

that additional revenue benefits residents: any surplus can be returned through the income tax, so a reform that raises revenue while holding every resident's utility fixed is Pareto-improving. Given this definition, the Pareto-efficient uniform commodity tax rate in this economy follows a simple inverse elasticity rule:

$$\frac{t}{1+t} = \frac{1}{\mathcal{E}^V} \quad (1)$$

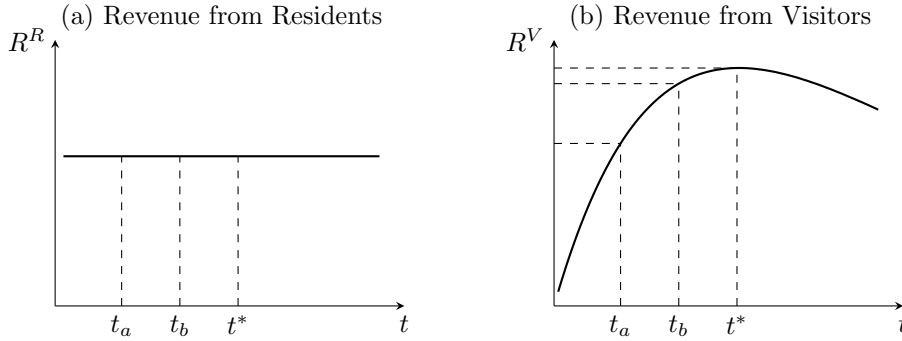
where $\mathcal{E}^V \equiv -\frac{1+t}{C_V} \frac{\partial C_V}{\partial t}$ is the elasticity of aggregate pre-tax spending by visitors in the economy. This elasticity includes both intensive and extensive margin responses by visitors.

The derivation follows from the basic insights of Feldstein [1999]: for any change in the commodity tax rate there is an income tax reform that leaves every resident's budget constraint unchanged. Starting from $(T(\cdot), t)$ and moving the commodity rate to $\hat{t}$, the reformed schedule

$$\hat{T}(z) = z - \frac{1+\hat{t}}{1+t} (z - T(z))$$

holds the purchasing power of every earnings level fixed, and therefore leaves residents' taxable income, consumption, welfare, and the revenue collected from them unchanged—only the income/commodity composition of that revenue shifts. Varying $\hat{t}$ thus traces out infinitely many tax systems that are welfare- and revenue-equivalent for residents but not revenue-equivalent for tourists. The Pareto efficient rate is the one that maximizes revenue from tourists (as per equation 1).

Figure 1: Revenue from Residents and Tourists under Different Tax Systems



Note: Panel (a) plots revenue collected from residents along the family of tax systems in which each commodity tax rate t is paired with the income tax adjustment that holds every resident's purchasing power constant. Along this family, resident welfare and resident revenue are invariant to t ; panel (b) shows that revenue from visitors is not.

Figure 1 illustrates this result. Note its generality: nothing has been assumed about resident utility functions or tourist demand.

This benchmark result can be restated in a useful way. From the perspective of residents, a uniform

commodity tax paired with its offsetting income tax adjustment is no tax at all: together, the pair amount to a destination-based consumption tax whose entire real burden falls on visitors—the one group of consumers the income tax cannot reach. Residents may remit the tax at purchase, but they bear no net burden from it.³

2 Optimal Differentiated Commodity Tax Rates

Differentiated commodity taxation requires more structure on behavior. For simplicity, I consider the case where there are two goods available for sale in the destination economy, both of which may be purchased by residents and visitors.⁴

Resident Demand Each resident has some type $w \in \mathcal{W}$. A type w resident chooses taxable income $z(w)$ and consumption of two commodities $x_1(w)$ and $x_2(w)$ to solve

$$\max_{z, x_1, x_2} \{u(\phi(x_1, x_2), z; w) : (1 + t_1)x_1 + (1 + t_2)x_2 = z - T(z)\}, \quad (2)$$

where t_i is the commodity tax rate applied to good $i \in \{1, 2\}$ and $T(\cdot)$ is a nonlinear income tax schedule. In what follows, it will often be helpful to reframe the resident's problem as a two-stage problem. In the second stage, the resident chooses commodity consumption conditional on their choice of taxable income z . Let $\Phi(z)$ represent the indirect utility the agent gets from this second stage problem for a given value of taxable income:

$$\Phi(z) \equiv \max_{x_1, x_2} \{\phi(x_1, x_2) : (1 + t_1)x_1 + (1 + t_2)x_2 = z - T(z)\}. \quad (3)$$

In the first stage, a type w resident chooses a value of taxable income $z(w)$ which maximizes their utility, anticipating how this choice impacts the value of the second stage problem:

$$\max_z \{u(\Phi(z), z; w)\}. \quad (4)$$

Because the second stage problem does not depend directly on w , commodity consumption is homogeneous across residents conditional on taxable income: I write $x_i(z)$ for the demand for good i of any resident with taxable income z , so a type w resident demands $x_i(z(w))$.

Given these assumptions, if the economy only consisted of residents, these preferences would imply that only

³The reason is simple: with a common wedge on all resident consumption, the offsetting income tax adjustment restores every resident's budget set exactly, so only visitors bear the tax. Appendix B states this result in a fully specified open economy; the equivalence holds to first order provided the consumption tax base extends to residents' consumption of traded goods.

⁴As discussed in the introduction, a full exploration of the multi-good case is beyond the scope of this paper.

a uniform commodity tax is efficient, as per the classic result of Atkinson and Stiglitz [1976]. However, this economy also includes a group of visitor consumers.

Visitor Demand Each (potential) visitor to the destination economy has some type $(\theta, \xi) \in \Theta \times \mathbb{R}$, where θ is a preference parameter that determines the visitor's demand behavior if they choose to visit the destination, and ξ is the value of the visitor's outside option (the value of not visiting the destination). If a visitor with type θ preferences chooses to visit the destination, their demand for each commodity in the destination economy ($x_1^V(\theta)$ and $x_2^V(\theta)$), and their total spending on goods outside the destination economy ($x_o^V(\theta)$) are chosen to solve

$$\max_{x_1, x_2, x_o} \{u(x_1, x_2, x_o; \theta) : (1 + t_1)x_1 + (1 + t_2)x_2 + x_o = m(\theta)\}, \quad (5)$$

where $m(\theta)$ is the income of a type θ visitor. A type (θ, ξ) visitor chooses to visit the destination economy if and only if

$$v(\theta) > \xi$$

where $v(\theta) \equiv u(x_1^V(\theta), x_2^V(\theta), x_o^V(\theta); \theta)$ is the indirect utility function associated with (5).⁵

The Outside Good The outside good x_o is a composite of all visitor consumption the destination cannot tax. Consider a prospective visitor who finds the tax-inclusive cost of a trip too high. One response is to visit anyway, but to spend less while traveling—to buy a luxury handbag from a retailer at home, say, rather than one in the destination. Responses of this kind, made conditional on visiting, are substitution toward x_o . A different response is to skip the trip altogether—to vacation somewhere else entirely, or to stay home and spend the money on home renovations. In the model this is a separate margin—the visit decision—introduced below. The two margins are distinct objects in the formal analysis, though from the standpoint of the destination's aggregate tax base both are leakage toward consumption it cannot tax, and section 2.3 shows they can be combined into a single aggregate leakage elasticity.

Tax Revenue Let the total number of residents in the economy be N_R , and the total number of (potential) visitors to the destination economy be N_V .⁶ Further, suppose that the distribution of resident types is

⁵Readers may wonder why residents have weakly separable preferences but tourists do not. The goal of this paper is to demonstrate the manner in which *tourist consumption alone* creates a rationale for differential commodity taxation. Relaxing weak separability for residents would create additional reasons to do so, for the purpose of tagging. Exploring the interactions between different motives for differentiation is beyond the scope of this paper. Moreover, altering the preference assumptions we have imposed for tourists would not change the results.

⁶In section 2.2, I consider the case where visitor arrivals are exogenous, so that N_V is the number of individuals who actually make a trip to the destination economy and consume some goods while there. More generally, N_V is the maximum possible number of individuals who might make such a trip.

described by F_W and the distribution of visitor preference parameters is described by G_Θ . Further, suppose that the conditional distribution of ξ is $\Pi(\cdot|\theta)$,⁷ so that the probability a type θ visitor chooses to visit the destination economy is

$$\Pr\{\xi < v(\theta) | \theta\} = \Pi(v(\theta) | \theta).$$

Total tax revenue in the destination economy is

$$R \equiv N_R \int T(z(w)) dF(w) + t_1(X_1^R + X_1^V) + t_2(X_2^R + X_2^V),$$

where total resident consumption of good i is

$$X_i^R \equiv N_R \int x_i(z(w)) dF(w),$$

and total visitor consumption of good i is

$$X_i^V \equiv N_V \int x_i^V(\theta) \Pi(v(\theta) | \theta) dG(\theta).$$

Throughout, producer prices are fixed (normalized to one) and the production side of the economy is left implicit: there are no profits, and commodity tax revenue is the only channel through which visitor demand affects the destination's resources. Appendix B shows that this is more than a convenient simplification. The model is, to first order, the reduced form of a diversified small open economy in which goods are produced from labor, residents also consume an importable good subject to a destination-based use tax, and the trade balance ties tourist spending to the economy's imports. Section 2.5 records the mapping; the substantive amendments to the formulas are that resident demand elasticities are read as conditional on resident demand for the traded good, and that tax wedges are read on consumer prices. Profits under imperfect competition—and the question of how they would be owned and taxed—are genuinely outside the model; I return to this in the conclusion.

2.1 Distribution Neutral Tax Reforms

To obtain a characterization of the Pareto efficient tax structure, we will consider tax reforms that jointly change these parameters in a manner that is distribution neutral for residents, in the spirit of Kaplow [2006].

⁷Allowing the distribution of ξ to vary with θ means the model is equivalent to one where a visitor chooses to visit the destination economy if and only if

$$u(x_1^V(\theta), x_2^V(\theta), x_o^V(\theta); \theta) > v_o(\theta, \xi)$$

where $v_o(\theta, \xi)$ is the indirect utility of not visiting the destination economy.

Suppose that the economy begins with the tax system $(t_1, t_2, T(\cdot))$. Now, consider a joint tax reform which marginally increases tax rate on good i by $dt_i > 0$ while also marginally changing income tax liability at every income level z by some amount $dT(z)$. The effect of this joint reform on the value of the residents' second stage problem (equation 3) is:

$$d\Phi(z) = -\frac{\partial\Phi(z)}{\partial I} [x_i(z) dt_i + dT(z)]$$

Notice, by setting $dT(z) = -x_i(z) dt_i$ for all z , we obtain a joint reform for which $d\Phi(z) = 0$ for all z .

This joint reform is distribution neutral: the income tax change compensates every resident for the lost purchasing power, so resident welfare is unchanged and the reform's welfare effect is entirely fiscal. Any distribution neutral reform that raises revenue is therefore Pareto-improving, and Pareto efficiency requires:⁸

$$\left. \frac{dR}{dt_1} \right|_{dT(z)=-x_1(z)dt_1} = 0 \quad \text{and} \quad \left. \frac{dR}{dt_2} \right|_{dT(z)=-x_2(z)dt_2} = 0 \quad (6)$$

Because $d\Phi(z) = 0$ at every z , residents' taxable income choices are also unchanged: income taxation can be ignored when evaluating the revenue effects of these reforms.

For simplicity, I assume visitor demand exhibits no income effects.⁹ Thus, the marginal revenue effect of a distribution neutral reform which increases the tax rate on good i is then

$$\begin{aligned} \left. \frac{dR}{dt_i} \right|_{dT(z)=-x_i(z)dt_i} &= X_i^V + N_R \underbrace{\int \left[t_i \frac{\partial x_i(z)}{\partial t_i} \Big|_{u,z=z(w)} + t_{-i} \frac{\partial x_{-i}(z)}{\partial t_i} \Big|_{u,z=z(w)} \right] dF(w)}_{\text{resident substitution effects}} \\ &\quad + N_V \underbrace{\int \left[t_i \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u + t_{-i} \frac{\partial x_{-i}^V(\theta)}{\partial t_i} \Big|_u \right] \Pi(v(\theta)|\theta) dG(\theta)}_{\text{visitor substitution effects}} \\ &\quad + N_V \underbrace{\int [t_i x_i^V(\theta) + t_{-i} x_{-i}^V(\theta)] \frac{d\Pi(v(\theta)|\theta)}{dt_i} dG(\theta)}_{\text{visitor extensive margin effects}} \end{aligned} \quad (7)$$

where $-i = \{1, 2\} \setminus i$, $\frac{\partial x_j(z)}{\partial t_k} \Big|_{u,z=z(w)}$ is the compensated response of resident demand for good j to the tax rate on good k (holding taxable income constant), $\frac{\partial x_j^V(\theta)}{\partial t_k} \Big|_u$ is the compensated response of visitor demand for good j to the tax rate on good k , and $\frac{d\Pi(v(\theta)|\theta)}{dt_k}$ is the effect of the tax rate on good k on the probability that a type θ visitor visits the destination economy. Notice, the revenue effect of this joint reform does not

⁸Throughout, I characterize interior optima, at which the relevant first-order conditions hold with equality. At a corner—a binding nonnegativity constraint on a tax rate, for instance—the corresponding condition is replaced by a one-sided inequality.

⁹If tourism expenditures are a small share of total expenditures for visitors, this assumption will be approximately correct. Nonetheless, the interested reader may refer to appendix A to see how visitor income effects can be incorporated into the model. They enter the optimal tax characterizations in a way that closely parallels the extensive margin responses.

include income effects for residents, as—by construction—the reform holds resident incomes constant.

2.2 A Corlett-Hague Rule for Tourist Taxation

Let us consider the case where there are no extensive margin responses. An optimal tax characterization can be obtained using an approach that parallels the standard derivation of the Corlett-Hague rule. Applying basic results from demand theory (see Appendix A), we can re-write the efficiency conditions (6) as:

$$-1 = \left(\frac{t_2}{1+t_2} - \frac{t_1}{1+t_1} \right) \mathcal{E}_1^V + \frac{t_2}{1+t_2} \mathcal{E}_{1,o}^V + \frac{X_1^R}{X_1^V} \left(\frac{t_2}{1+t_2} - \frac{t_1}{1+t_1} \right) \mathcal{E}_{1|z}^R \quad (8)$$

$$-1 = \left(\frac{t_1}{1+t_1} - \frac{t_2}{1+t_2} \right) \mathcal{E}_2^V + \frac{t_1}{1+t_1} \mathcal{E}_{2,o}^V + \frac{X_2^R}{X_2^V} \left(\frac{t_1}{1+t_1} - \frac{t_2}{1+t_2} \right) \mathcal{E}_{2|z}^R \quad (9)$$

where

$$\mathcal{E}_i^V \equiv \int \frac{x_i^V(\theta)}{\frac{1}{N_V} X_i^V} \left(-\frac{1+t_i}{x_i^V(\theta)} \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u \right) dG(\theta)$$

is the aggregate (compensated) elasticity of visitor demand for good i ,

$$\mathcal{E}_{i|z}^R \equiv \int \frac{x_i(z(w))}{\frac{1}{N_R} X_i^R} \left(-\frac{1+t_i}{x_i(z(w))} \frac{\partial x_i(z(w))}{\partial t_i} \Big|_u \right) dF(w)$$

is the aggregate (compensated) elasticity of resident demand for good i , and

$$\mathcal{E}_{i,o}^V \equiv \int \frac{x_i^V(\theta)}{\frac{1}{N_V} X_i^V} \left(-\frac{1}{x_i^V(\theta)} \frac{\partial x_i^V(\theta)}{\partial p_o} \Big|_u \right) dG(\theta)$$

is the aggregate (compensated) cross-elasticity of visitor demand for good i with respect to the price of the outside option (i.e., consumption outside the destination economy). Note the sign convention: with the leading minus sign, $\mathcal{E}_{i,o}^V$ is *negative* when good i and outside consumption are Hicksian substitutes and *positive* when they are Hicksian complements, with stronger substitutability making it more negative. Thus statements below of the form “good 2 is more substitutable for outside consumption than good 1” correspond to $\mathcal{E}_{1,o}^V > \mathcal{E}_{2,o}^V$.

Combining equations (8) and (9) we obtain

$$\frac{\frac{t_1}{1+t_1}}{\frac{t_2}{1+t_2}} = \frac{\mathcal{E}_{1,o}^V + \overbrace{\left[\mathcal{E}_1^V + \mathcal{E}_2^V + \left(\frac{X_1^R}{X_1^V} \right) \mathcal{E}_{1|z}^R + \left(\frac{X_2^R}{X_2^V} \right) \mathcal{E}_{2|z}^R \right]}^{\text{attenuation factor}}}{\mathcal{E}_{2,o}^V + \underbrace{\left[\mathcal{E}_1^V + \mathcal{E}_2^V + \left(\frac{X_1^R}{X_1^V} \right) \mathcal{E}_{1|z}^R + \left(\frac{X_2^R}{X_2^V} \right) \mathcal{E}_{2|z}^R \right]}_{\text{attenuation factor}}} \quad (10)$$

This characterization closely parallels the classic Corlett-Hague rule. Both the numerator and the denominator of equation (10) are a sum of five components, but four of these are identical (what I have labeled the *attenuation factor*). Equation (10) shows that differential commodity taxation is only efficient if aggregate demand for one of the two commodities is more responsive to the price of consumption outside the destination economy. That is to say, the tax rate on good 2 should be lower than the rate on good 1 if and only if good 2 is more substitutable for outside consumption than good 1: $\mathcal{E}_{1,o}^V > \mathcal{E}_{2,o}^V$.¹⁰

The responses in the attenuation factor determine neither whether differentiation occurs nor its direction—only its magnitude: the larger the attenuation factor, the smaller the efficient differentiation. The more responsive visitor and resident demand are to own-prices, the more attenuated the differentiation. Note as well that equation (10) provides no rationale for taxing goods with relatively high tourist demand more heavily; a high resident share of consumption only attenuates differentiation further.

Spending that visitors find easier to substitute for purchases outside the destination economy makes for a relatively unattractive tax base, as this expenditure can more easily escape taxation if rates rise. Consequently, all else equal, lower tax rates should be placed on such purchases. However, the resulting tax rate differentiation generates some distortions in consumption patterns of both residents and visitors. The attenuation factor includes the own-price elasticities of demand for both visitors and residents as a measure of the fiscal externalities these distortions generate. The larger these terms are, the greater the revenue loss caused by the distortionary effects of differential commodity taxation. Furthermore, the larger the resident share of consumption is, the greater the revenue loss caused by resident behavioral responses is relative to the revenue gains generated by more targeted taxation of tourists.¹¹ If too much rate differentiation occurs, these revenue losses will dominate the revenue gains generated by targeting commodity taxes towards spending which cannot easily escape taxation. Efficient differentiation must balance these two effects.

The parallel to Corlett and Hague [1953] is striking. In that paper, without a nonlinear income tax, differentiation is optimal only when the untaxed good—leisure—is more substitutable for some commodities than others, and own-price responses determine only the degree of differentiation. Here the untaxed good is visitors' outside consumption, but the logic is the same.

¹⁰This ranking presumes positive tax rates and that both the numerator and denominator of equation (10) are positive. Using the adding-up condition for visitor compensated demands, the numerator can be written as $\mathcal{E}_2^V - \mathcal{E}_{12}^V + \left(\frac{X_1^R}{X_1^V}\right) \mathcal{E}_{1|z}^R + \left(\frac{X_2^R}{X_2^V}\right) \mathcal{E}_{2|z}^R$, where $\mathcal{E}_{12}^V$ is the aggregate compensated cross-elasticity of visitor demand for good 1 with respect to the price of good 2 (negative for Hicksian substitutes), and analogously for the denominator. Both are therefore positive whenever the two destination goods are not strong Hicksian complements for visitors.

¹¹Note, in the limit, as the tourist share of consumption vanishes, the attenuation factor tends toward infinity, driving the ratio of tax rates in equation (10) toward unity; back to the Atkinson-Stiglitz benchmark.

2.3 Endogenous Arrivals

Incorporating endogenous arrivals provides an additional rationale for tax rate differentiation. Let

$$\bar{N}_V \equiv N_V \int \Pi(v(\theta) | \theta) dG(\theta),$$

be the mass of realized arrivals, and let $\bar{x}_i^V \equiv X_i^V / \bar{N}_V$ be mean consumption of good i among these arrivals. In Appendix A, I show that including these responses leads to the following result regarding efficient relative tax rates:¹²

$$\frac{\frac{t_1}{1+t_1}}{\frac{t_2}{1+t_2}} = \frac{\mathcal{A} + \mathcal{E}_{1,o}^V + \text{Cov}\left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \alpha_2(\theta) \epsilon_{ext}^V(\theta)\right)}{\mathcal{A} + \mathcal{E}_{2,o}^V + \text{Cov}\left(\frac{x_1^V(\theta)}{\bar{x}_1^V} - \frac{x_2^V(\theta)}{\bar{x}_2^V}, \alpha_1(\theta) \epsilon_{ext}^V(\theta)\right)}, \quad (11)$$

where $\mathcal{A}$ denotes the attenuation factor of equation (10), the elasticity of the probability of visiting with respect to the total cost of a visit is

$$\epsilon_{ext}^V(\theta) \equiv \frac{(1+t_1)x_1^V(\theta) + (1+t_2)x_2^V(\theta)}{v(\theta)} \frac{\partial v(\theta)}{\partial m(\theta)} \frac{v(\theta) \Pi'(v(\theta) | \theta)}{\Pi(v(\theta) | \theta)},$$

and

$$\alpha_i(\theta) \equiv \frac{(1+t_i)x_i^V(\theta)}{(1+t_1)x_1^V(\theta) + (1+t_2)x_2^V(\theta)},$$

is the share of type θ 's tax-inclusive destination spending devoted to good i .

Both the numerator and the denominator of the right hand side of equation (11) now contain a covariance term measuring a good-specific endogenous arrival effect.¹³ To see what these terms do, consider the simple case where $\mathcal{E}_{1,o}^V = \mathcal{E}_{2,o}^V$, so that substitution provides no reason for differentiation. Because $\alpha_1(\theta) + \alpha_2(\theta) = 1$, what remains depends only on which good's consumption has the algebraically larger covariance with the extensive margin elasticity: presuming, as with equation (10), positive rates and a positive numerator and denominator, efficiency requires a higher tax rate on good 1 whenever

$$\text{Cov}\left(\frac{x_2^V(\theta)}{\bar{x}_2^V}, \epsilon_{ext}^V(\theta)\right) > \text{Cov}\left(\frac{x_1^V(\theta)}{\bar{x}_1^V}, \epsilon_{ext}^V(\theta)\right).$$

The extensive margin pushes the efficient rate down on the good disproportionately consumed by the visitors most inclined to forgo a trip as its cost rises. Raising the tax on such a good deters its own most intensive

¹²Note, in this version of the model, all expectations (including aggregate elasticities) and covariances are taken over the distribution of realized arrivals.

¹³The product $\alpha_i(\theta) \epsilon_{ext}^V(\theta) = -\frac{\partial \log \Pi(v(\theta) | \theta)}{\partial \log(1+t_i)}$ is the elasticity of type θ 's probability of visiting with respect to the tax-inclusive price of good i .

consumers, and the revenue they would have contributed leaves with them.¹⁴

One way of viewing this result is as a tagging rationale in the spirit of Akerlof [1978]. If the composition of destination spending provides information about visitors' extensive margin responsiveness, the planner can exploit that information to shift the burden of visitor taxation onto those least responsive on the extensive margin.

To build further intuition for the extensive margin terms, consider two possible types of tourists: budget-conscious visitors and luxury travelers. Suppose good 1 is an aggregation of "luxury consumption" (broadly defined) and is disproportionately consumed by luxury tourists, while good 2 is more basic forms of consumption. All else equal, does the extensive margin favor higher taxes on luxury consumption? If budget-conscious visitors are the more elastic group at the extensive margin, the answer is yes. In such a case, levying higher taxes luxury goods is relatively attractive because it shifts the burden of taxation away from this highly responsive group.¹⁵

Figure 2 illustrates the possible configurations. In panels (a) and (b), consumption of good 2 has the algebraically larger covariance with the extensive margin elasticity—budget-conscious visitors are the more easily deterred group—and the extensive margin favors a higher rate on good 1. Panel (c) reverses this configuration. In panel (d), arrival responsiveness is unrelated to spending composition, and the extensive margin provides no rationale for differentiation.

The covariance terms also admit an aggregate demand interpretation. Since X_i^V assigns zero destination purchases to potential visitors who do not come, its price responses already incorporate both changes in purchases by continuing visitors and changes in the number and composition of arrivals. Define

$$\hat{\mathcal{E}}_i^V \equiv -\frac{1+t_i}{X_i^V} \frac{\partial X_i^V}{\partial t_i}, \quad \hat{\mathcal{E}}_{i,o}^V \equiv \frac{1}{X_i^V} \left[(1+t_1) \frac{\partial X_i^V}{\partial t_1} + (1+t_2) \frac{\partial X_i^V}{\partial t_2} \right],$$

the total own-price elasticity of aggregate visitor demand for good i and the total response of that demand to a proportional increase in all destination prices. The second of these is negative when the tax base for good i escapes the destination as it becomes uniformly more expensive, whether through substitution toward consumption elsewhere by continuing visitors or through nonarrival, and is more negative the more readily the base escapes. With arrivals exogenous these reduce to the elasticities of section 2.2: $\hat{\mathcal{E}}_i^V = \mathcal{E}_i^V$ immediately, and $\hat{\mathcal{E}}_{i,o}^V = \mathcal{E}_{i,o}^V$ by visitor adding-up.

¹⁴Appendix A derives equation (11) from the type-level efficiency conditions and records an equivalent representation in which the extensive margin enters through a single covariance.

¹⁵If we allowed for non-zero visitor income effects, they would play a parallel role. Appendix A shows that such effects favor a lower rate on any good which is disproportionately consumed by visitors with high marginal propensities to spend on destination goods. In practice, which commodities one should expect to proxy for a high marginal propensity to spend in the destination is unclear.

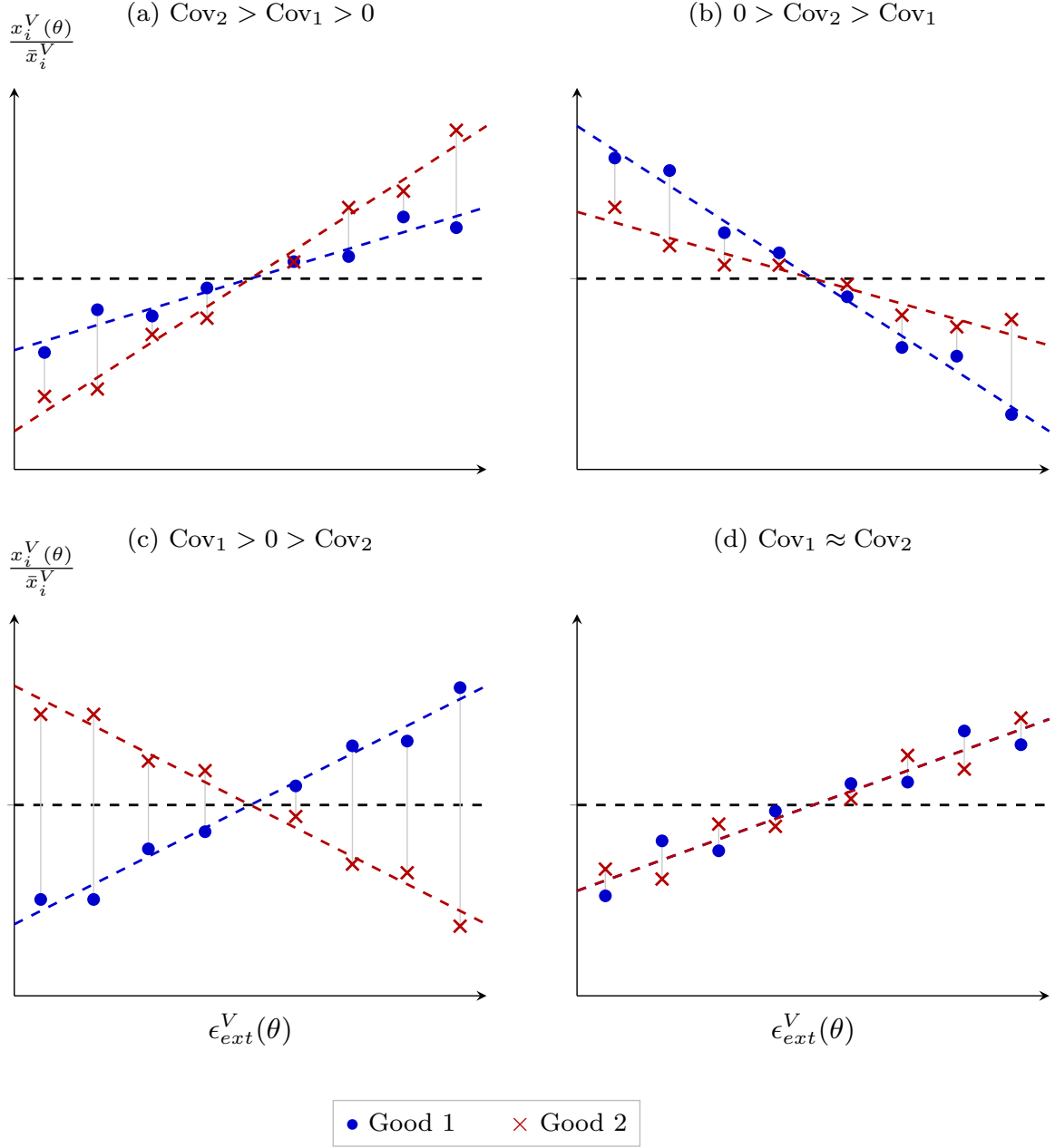


Figure 2: Extensive margin selection and optimal rate differentiation. Each panel plots relative consumption intensity $x_i^V(\theta)/\bar{x}_i^V$ against the extensive margin elasticity $\epsilon_{ext}^V(\theta)$ for goods 1 (blue circles) and 2 (red crosses). Dashed lines show OLS regression fits, whose slopes are proportional to $\text{Cov}_i \equiv \text{Cov}(x_i^V(\theta)/\bar{x}_i^V, \epsilon_{ext}^V(\theta))$. The extensive margin favors a lower tax rate on the good whose fitted line has the algebraically larger slope (presuming positive rates and a positive numerator and denominator in equation (11)): visitors who are relatively intensive consumers of that good are disproportionately likely to abandon the destination when prices rise. In panels (a) and (b) the extensive margin favors $t_1 > t_2$; in panel (c) it favors $t_2 > t_1$; in panel (d) equal slopes imply no extensive-margin force for rate differentiation.

These two elasticities absorb the arrival response in different amounts, and the covariance terms record the difference. Raising t_i deters arrivals in proportion to $\alpha_i(\theta)\epsilon_{ext}^V(\theta)$, so $\hat{\mathcal{E}}_i^V$ picks up only the arrivals

deterred by good i 's own price. A proportional increase in both prices deters arrivals through each price, so $\hat{\mathcal{E}}_{i,o}^V$ picks up the whole arrival response. Now sum $\hat{\mathcal{E}}_{1,o}^V$, $\hat{\mathcal{E}}_1^V$, and $\hat{\mathcal{E}}_2^V$, splitting the arrival leakage with $\alpha_1(\theta) + \alpha_2(\theta) = 1$. The own-price absorption for good 1 cancels against part of the leakage, leaving

$$\hat{\mathcal{E}}_{1,o}^V + \hat{\mathcal{E}}_1^V + \hat{\mathcal{E}}_2^V = \mathcal{E}_{1,o}^V + \mathcal{E}_1^V + \mathcal{E}_2^V + \text{Cov}\left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \alpha_2(\theta) \epsilon_{ext}^V(\theta)\right),$$

and analogously with the indices interchanged. Under the maintained assumption of no visitor income effects, the arrival response preserves the symmetry of the aggregate visitor demand system, and the optimality condition can be written as an ordinary Corlett-Hague rule—equation (10) itself—with each visitor elasticity replaced by its arrivals-inclusive counterpart. Note, the correspondence holds bracket by bracket and not term by term. The covariances are not arrival corrections to $\mathcal{E}_{i,o}^V$. They are what is left over once the arrival responses have been divided between the attenuation factor and the leakage elasticities. This parallels the uniform tax case, where the elasticity $\mathcal{E}^V$ of section 1 already included both intensive and extensive margin responses.

2.4 The Efficient Level of Taxation

Combining the efficiency conditions in equation (6), we have that efficient commodity taxation must satisfy

$$(1 + t_1) \left. \frac{dR}{dt_1} \right|_{dT(z)=-x_1(z)dt_1} + (1 + t_2) \left. \frac{dR}{dt_2} \right|_{dT(z)=-x_2(z)dt_2} = 0.$$

This condition can be re-arranged to obtain the following, more intuitive presentation (see Appendix A):

$$\frac{t_1 X_1^V + t_2 X_2^V}{(1 + t_1) X_1^V + (1 + t_2) X_2^V} = \frac{1}{-\frac{t_1 X_1^V}{t_1 X_1^V + t_2 X_2^V} \hat{\mathcal{E}}_{1,o}^V - \frac{t_2 X_2^V}{t_1 X_1^V + t_2 X_2^V} \hat{\mathcal{E}}_{2,o}^V} \quad (12)$$

where $\hat{\mathcal{E}}_{i,o}^V$ is the aggregate leakage elasticity of section 2.3.

The left hand side of equation (12) is commodity taxes paid by visitors as a share of their total spending in the destination—the efficient average commodity tax rate visitors face. As in the uniform case, the efficient level follows an inverse elasticity rule. The elasticity in question is a revenue-weighted average of the rates at which the two tax bases leak out of the destination when all destination prices rise together. Like the elasticity $\mathcal{E}^V$ of section 1, it counts both the substitution of continuing visitors toward consumption elsewhere and the loss of visitors who do not come at all, so the endogeneity of arrivals requires no separate term.

Like equation (1), equation (12) characterizes the efficient level of visitor taxation only in terms of sufficient

statistics that pertain to visitor commodity demand. This generalizes a key result from the uniform tax case: the efficient level of commodity taxation is one which maximizes revenue extracted from tourists.¹⁶

2.5 A Small-Open-Economy Reinterpretation

The results above were derived holding producer prices fixed, with no production side and no trade. In this section I close the model as a small open economy in which producer prices and the wage are endogenous. Notably, doing so changes very little: the characterizations above continue to describe Pareto efficient taxation, subject to two amendments. Appendix B contains the derivations.

Goods 1 and 2 remain nontraded, produced from resident labor—the economy’s only primary factor—under perfect competition with constant returns to scale. A third good, good 0, is traded at a fixed world price and serves as the numéraire. Visitors do not purchase it within the destination; residents’ use of it—whether produced domestically or imported—carries a destination-based use tax t_0 . Because marginal labor requirements vary with the composition of output, tourist demand now moves both nontraded producer prices and the wage; the fixed-price environment above is the special case of linear technology. I assume production remains diversified, so that all three goods continue to be produced, and tourist spending finances the economy’s net imports through the current account.

The first amendment is that each resident elasticity $\mathcal{E}_{i|z}^R$ is read as its counterpart conditional on resident demand for the traded good,

$$\tilde{\mathcal{E}}_{i|z}^R \equiv \mathcal{E}_{i|z}^R - \frac{\mathcal{E}_{i0|z}^R \mathcal{E}_{0i|z}^R}{\mathcal{E}_{00|z}^R},$$

where $\mathcal{E}_{ik|z}^R$ is the aggregate compensated cross-elasticity of resident demand for good i with respect to the consumer price of good k , as defined in Appendix B. Throughout, the compensated own-price response for the traded good is assumed strictly negative, so $\mathcal{E}_{00|z}^R > 0$. Wedges are likewise read on consumer prices: each $\frac{t_i}{1+t_i}$ becomes $\frac{t_i}{q_i}$, where q_i is the consumer price of good i . With producer prices fixed at one, the two coincide. More generally, wherever a tax-inclusive price $1 + t_i$ appears in the formulas—in the visitor spending shares and the extensive margin elasticity, for instance—it is read as q_i .

The second amendment is to the compensating reform. With the wage endogenous, returning to residents their first-order loss in purchasing power is no longer enough: the income tax reform must also replicate the induced movements in prices and the wage, leaving labor supply—not merely utility—undisturbed. This is feasible precisely because labor is the only primary factor, so a nonlinear income tax can span the one factor

¹⁶Because the derivatives in $\hat{\mathcal{E}}_{i,o}^V$ are total responses of aggregate visitor demand, visitor income effects need no separate treatment here: they are already counted. Appendix A derives equation (12) and records the explicit decomposition of the leakage elasticity into substitution, arrival, and income components, equation (23).

price that moves [Jacquet and Lehmann, 2025]. Given these amendments, the Corlett-Hague rule of section 2.2, the tagging results of section 2.3, and the efficient level of section 2.4 all hold in the same form as before, with every visitor-side object unchanged.¹⁷

The visitor side of the problem is unaffected because visitors are simply not involved in the eliminated margin: they do not buy good 0 in the destination, and they do not condition their travel decisions on its price. Moreover, the use-tax base does not move along the compensated reforms. A companion adjustment to the price of good 0 holds aggregate resident demand for good 0 fixed, while the income tax adjustment compensates residents and neutralizes the induced wage movement (Appendix B). The arrival-selection covariances still matter less as the attenuation factor grows, but no new visitor-side sufficient statistic appears—the resident side now requires the cross-elasticities with the traded good that enter $\tilde{\mathcal{E}}_{i|z}^R$. General equilibrium determines where the sufficient statistics are evaluated and how the tax wedges are implemented, but it does not provide any additional reason to differentiate rates.

The conditional elasticities have a familiar interpretation [Pollak, 1969, Neary and Roberts, 1980]: $\tilde{\mathcal{E}}_{i|z}^R$ is the compensated response of aggregate resident demand for good i when the price of good 0 adjusts to hold aggregate resident demand for good 0 fixed. This is an as-if representation—it does not assert that consumption of good 0 is literally frozen along equilibrium tax changes—but it delivers a Le Chatelier bound [Samuelson, 1947]: fixing a quantity margin weakly shrinks substitution possibilities, so $\tilde{\mathcal{E}}_{i|z}^R \leq \mathcal{E}_{i|z}^R$. The resident component of the attenuation factor is thus weakly smaller, and every visitor-side motive for differentiation correspondingly less attenuated. Note as well the more substantive lesson: what an enforceable destination-based tax can reach depends on where consumption occurs. Visitors’ outside consumption lies beyond reach because it occurs abroad, while residents’ consumption of good 0 occurs at home, and so falls within the tax base wherever the good was produced.

The use tax supporting this reduction satisfies

$$\frac{t_0}{q_0} = -\frac{\mathcal{E}_{01|z}^R \frac{t_1}{q_1} + \mathcal{E}_{02|z}^R \frac{t_2}{q_2}}{\mathcal{E}_{00|z}^R}. \quad (13)$$

Note, when good 0 is a net substitute for both home goods, the use-tax wedge is a weighted average of the two home-good wedges, and so lies between them; when the home goods carry a common wedge, resident adding-up gives exactly that wedge on good 0. It may not be obvious that good 0 should be taxed at all,

¹⁷With producer prices endogenous, “revenue from tourists” requires one qualification. A tax change now moves producer prices, so the change in statutory visitor receipts differs from the visitor block of the revenue ledger in Appendix B by the revaluation term $X_1^V dp_1 + X_2^V dp_2$: visitor spending at higher producer prices is additional resident factor income, which the compensating income tax reform recaptures. What equation (12) maximizes in the open economy is this total fiscal surplus from visitor spending—commodity receipts plus recaptured factor income—rather than statutory commodity receipts alone. The formula itself is unchanged.

since only residents consume it. But where good 0 substitutes for the home goods, leaving it untaxed while goods 1 and 2 carry positive rates would distort resident consumption toward it; and since residents can be compensated through the income tax, the planner prefers to remove that distortion and compensate them for the loss. Levying the tax on a destination basis also leaves the optimum production efficient: every wedge falls on final use, so no producer margin is distorted, as an origin-based tax on the traded good would be. This is consistent with the prescription of Diamond and Mirrlees [1971] and with Jacquet and Lehmann [2025].¹⁸

If the use tax is unavailable or unenforceable, resident substitution toward good 0 enters the differentiation rule directly, in the same algebraic position as visitors' substitution toward outside consumption. A consumption tax base broad enough to reach traded goods is thus a precondition for the clean tourist-only differentiation rule derived above.

3 Conclusion

This paper characterizes optimal commodity taxation in the presence of tourists when nonlinear income taxes are also available—the policy-relevant case for any destination whose government levies its own income tax, whether national or subnational. Notably, my results show that efficient rate differentiation is at odds with the intuitive idea that higher tax rates should be levied on commodities disproportionately demanded by tourists. Rather, tax rates should be higher on goods which visitors can less readily substitute for outside consumption and on goods which act as tags for visitors with unresponsive arrival decisions.

These results also have a trade-theoretic interpretation. Tourism is the export of nontraded services—the consumer, not the good, crosses the border—and tourist receipts ultimately finance the destination's net imports and public revenue (Section 2.5). Seen this way, the tourist-facing component of the tax system characterized here is the destination's export pricing, subject to the constraint that visitors and residents face the same prices. In the benchmark, lower rates apply to the nontraded services whose foreign buyers can most easily redirect their spending outside the destination; the extensive margin terms of section 2.3 can reinforce or overturn this ranking.

What might the concrete implications of these results be? Suppose policymakers in a destination economy are considering placing a higher tax rate on luxury goods, on the basis that tourists account for a disproportionate

¹⁸The compensating adjustment is mechanically budget-neutral: at fixed quantities, the direct use-tax revenue exactly finances the offset. Equation (13) then pins down the interior t_0 at which the marginal revenue effect of behavioral substitution vanishes. Residents suffer no first-order welfare loss along this compensated comparison, though their consumption bundle is only left fully undistorted when commodity taxation is uniform.

share of demand for them. The results of this paper suggest the opposite policy may be desirable: luxury purchases are readily substitutable for luxury goods purchased in other jurisdictions, suggesting a lower tax. Extensive margin responses could reinforce or undo this prescription. What matters is whether luxury purchases tag visitors who are relatively likely to abandon the destination as its cost rises. If luxury travelers can readily substitute toward competing destinations, luxury goods tag exit-prone visitors and the case for a lower rate is reinforced; if budget-conscious visitors are the more marginal group, it is undone. That big spenders generate more tax revenue conditional on visiting matters directly for the efficient level of visitor taxation; it creates no independent force for differentiation, though spending levels do shape the strength of the tagging relationships and so can affect whether they overturn the substitution-based prescription.

Alternatively, consider placing a relatively low tax rate on groceries. The results of this paper do not support doing so on the grounds that tourists make up a small share of grocery demand. If anything, the opposite holds: visitors generally cannot substitute food purchased in the destination for food purchased at home, suggesting a higher rate is warranted. Here too the extensive margin cuts in an ambiguous direction: if tourists who spend heavily on groceries are budget-conscious visitors whose trips are sensitive to total cost, grocery purchases tag exit-prone visitors and a lower rate regains appeal. A high grocery budget share alone need not generate a strong tagging force, however, if such visitors spend little in the destination overall.

This contrasts with existing results, derived in Ramsey frameworks without nonlinear income taxes, which favor taxing more highly the goods where tourist demand is relatively high [Hämäläinen, 2004, Gooroochurn, 2009, Kawahara, 2025]; those prescriptions are pertinent when the income tax on residents is absent, or insufficiently flexible to implement the compensating reforms studied here. The analysis also connects tourist taxation to recent characterizations of Pareto-efficient rate differentiation built, as here, on distribution-neutral reforms in the spirit of Kaplow [2006].

This connection suggests several directions for future extension of the results contained in this paper. There are many other documented reasons that optimal rates might be differentiated, including preference heterogeneity [Saez, 2002, Ferey et al., 2022], evasion/avoidance, credit constraints, and non-takeup of tax credits. Understanding how these different motives interact with the planner’s desire to extract revenue from tourists may be important to determining the policy implications of these results. The instrument set could also be enriched. I have restricted attention to linear commodity taxes; a fixed visitor-specific charge—an entry or departure fee—would turn the problem into the design of a two-part tariff, changing both the efficient level of commodity taxation and the fiscal mechanics of extensive margin tagging. Extending the analysis to many taxable goods is another natural step; as noted in the introduction, ranking tax rates in that setting requires confronting the full pattern of substitution across goods.

More importantly, extending these results to a richer general equilibrium setting is likely critical if the goal is to provide a framework for applied policy analysis. Section 2.5 takes the first step, closing the model as a small open economy in which the paper’s characterizations continue to hold with resident elasticities read as conditional on resident demand for the traded good (section 2.5). Two substantive extensions remain. The first concerns wages. The closure lets the wage move, but every sector employs the same homogeneous labor, so a single income-tax adjustment can neutralize its movement—the case in which general equilibrium is simplest. If commodity taxes instead move wages differentially across worker types, then unless the planner can levy a different schedule on every type, optimal tax characterizations must account for the fiscal externalities of wage changes and their predistributive consequences [Naito, 1999, Jacquet and Lehmann, 2025].¹⁹ The second concerns profits. If the sectors serving tourists earn rents under imperfect competition, the ownership and taxation of those rents can no longer be set aside, and the production side requires explicit cost primitives. While some general equilibrium analyses of tourism taxation exist [Gooroochurn and Sinclair, 2008, Kawahara, 2025], these do not account for nonlinear income taxation, leaving substantial scope for further development.

Accounting for congestion and environmental externalities offers another direction. Such externalities are likely partly non-atmospheric [Micheletto, 2008]: marginal damage varies across tourist types, and much of it—congestion at attractions, on roads, beaches, and trails—is unpriced, so taxes on hotels or restaurants are only indirect proxies for the activities that generate damage. Optimal commodity taxation in that setting involves not only discouraging some visits, but taxing relatively more highly the goods whose consumption tags tourists who impose large externalities relative to the revenue they provide.

Nonetheless, this paper provides a foundation on which a more complete guide to taxation in tourism-dependent economies can be built—and, because its characterizations are stated in estimable sufficient statistics, clear guidance on which empirical evidence on tourist demand can (or cannot) inform tax policy.

¹⁹Note, under the diversified closure the pass-through of statutory taxes to consumer prices is endogenous, and the optimal wedges are stated directly on consumer prices. Residents face those prices too, but the distribution-neutral income tax adjustment compensates them for the burden, so the net incidence falls on tourists. In a fully specialized economy, some of the incidence can instead be shifted onto wages. The empirical evidence suggests forward shifting is substantial, though not complete. For example, Fujii et al. [1985] find that hotel room taxes in Hawaii are “highly exportable,” with roughly two-thirds of the burden falling on tourists rather than local factors.

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A Derivation of Optimal Tax Characterization

This appendix derives the optimal tax characterizations for the general two good partial equilibrium model, including extensive margin and income effects that were excluded from the derivation in the main text. The full optimality condition $\left. \frac{dR}{dt_i} \right|_{dT(z)=-x_i(z)dt_i} = 0$ can be written as

$$\begin{aligned}
0 = & X_i^V + N_R \int \left[t_i \frac{\partial x_i(z)}{\partial t_i} \Big|_{u,z=z(w)} + t_{-i} \frac{\partial x_{-i}(z)}{\partial t_i} \Big|_{u,z=z(w)} \right] dF(w) \\
& + N_V \int \left[t_i \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u + t_{-i} \frac{\partial x_{-i}^V(\theta)}{\partial t_i} \Big|_u \right] \Pi(v(\theta)|\theta) dG(\theta) \\
& + N_V \int [t_1 x_1^V(\theta) + t_2 x_2^V(\theta)] \frac{d\Pi(v(\theta)|\theta)}{dt_i} dG(\theta) \\
& - N_V \int x_i^V(\theta) \left[t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right] \Pi(v(\theta)|\theta) dG(\theta)
\end{aligned} \tag{14}$$

Applying standard properties of Hicksian demand functions to the second stage resident compensated responses, we have

$$(1+t_i) \frac{\partial x_i(z)}{\partial t_i} \Big|_{u,z} + (1+t_{-i}) \frac{\partial x_{-i}(z)}{\partial t_{-i}} \Big|_{u,z} = 0.$$

Combined with Slutsky symmetry, this implies that

$$\frac{\partial x_{-i}(z)}{\partial t_i} \Big|_{u,z} = -\frac{1+t_i}{1+t_{-i}} \frac{\partial x_i(z)}{\partial t_i} \Big|_{u,z}. \tag{15}$$

Similarly, visitor compensated responses satisfy

$$(1+t_i) \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u + (1+t_{-i}) \frac{\partial x_{-i}^V(\theta)}{\partial t_{-i}} \Big|_u + \frac{\partial x_i^V(\theta)}{\partial p_o} \Big|_u = 0$$

where p_o is the price of (untaxed) consumption outside the tourist destination. Together with Slutsky symmetry, this gives us

$$\frac{\partial x_{-i}^V(\theta)}{\partial t_i} \Big|_u = -\frac{1+t_i}{1+t_{-i}} \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u - \frac{1}{1+t_{-i}} \frac{\partial x_i^V(\theta)}{\partial p_o} \Big|_u. \tag{16}$$

Recall that, for type θ visitors $\alpha_i(\theta)$ is the share of their destination spending that was allocated to good i and $\epsilon_{ext}^V(\theta)$ is their extensive margin elasticity (as defined in section 2.3). Thus

$$\frac{d\Pi(v(\theta)|\theta)}{dt_i} = -\frac{\alpha_i(\theta) \epsilon_{ext}^V(\theta)}{1+t_i} \Pi(v(\theta)|\theta), \tag{17}$$

and the extensive margin revenue term is

$$[t_1 x_1^V(\theta) + t_2 x_2^V(\theta)] \frac{d\Pi(v(\theta)|\theta)}{dt_i} = -x_i^V(\theta) \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \Pi(v(\theta)|\theta),$$

where $\bar{\tau}(\theta) \equiv \frac{t_1 x_1^V(\theta) + t_2 x_2^V(\theta)}{(1+t_1)x_1^V(\theta) + (1+t_2)x_2^V(\theta)}$ is the share of type θ 's destination spending collected as tax revenue, so that $\bar{\tau}(\theta) \epsilon_{ext}^V(\theta)$ is the fiscal externality associated with type θ 's nonarrival.

Using these equalities, equation 14 can be re-written as

$$\begin{aligned} 0 = & X_i^V + \left(\frac{t_{-i}}{1+t_{-i}} - \frac{t_i}{1+t_i} \right) N_V \int x_i^V(\theta) \varepsilon_i^V(\theta) \Pi(v(\theta)|\theta) dG(\theta) \\ & + \frac{t_{-i}}{1+t_{-i}} N_V \int x_i^V(\theta) \varepsilon_{i,o}^V(\theta) \Pi(v(\theta)|\theta) dG(\theta) \\ & + \left(\frac{t_{-i}}{1+t_{-i}} - \frac{t_i}{1+t_i} \right) N_R \int x_i(z(w)) \varepsilon_{i|z}^R(z(w)) dF(w) \\ & - N_V \int x_i^V(\theta) \left(t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right) \Pi(v(\theta)|\theta) dG(\theta) \\ & - N_V \int x_i^V(\theta) \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \Pi(v(\theta)|\theta) dG(\theta) \end{aligned}$$

where

$$\varepsilon_{i|z}^R(z) \equiv - \frac{1+t_i}{x_i(z)} \frac{\partial x_i(z)}{\partial t_i} \Big|_{u,z}$$

is the compensated elasticity of resident demand for good i (holding taxable income constant),

$$\varepsilon_i^V(\theta) \equiv - \frac{1+t_i}{x_i^V(\theta)} \frac{\partial x_i^V(\theta)}{\partial t_i} \Big|_u$$

is the compensated elasticity of visitor demand for good i , and

$$\varepsilon_{i,o}^V(\theta) \equiv - \frac{1}{x_i^V(\theta)} \frac{\partial x_i^V(\theta)}{\partial p_o} \Big|_u$$

is the cross-elasticity of visitor demand for good i with respect to the price of consumption outside the destination economy.

To aggregate, let $\bar{N}_V$, $\bar{x}_i^V$, and the realized-arrivals measure be as defined in Section 2.3. As noted there, with endogenous arrivals the aggregate elasticities are defined over realized arrivals:

$$\mathcal{E}_i^V \equiv \frac{N_V}{\bar{X}_i^V} \int x_i^V(\theta) \varepsilon_i^V(\theta) \Pi(v(\theta)|\theta) dG(\theta), \quad \mathcal{E}_{i,o}^V \equiv \frac{N_V}{\bar{X}_i^V} \int x_i^V(\theta) \varepsilon_{i,o}^V(\theta) \Pi(v(\theta)|\theta) dG(\theta),$$

which reduce to the definitions of section 2.2 when $\Pi \equiv 1$.

For any type-level quantity $\psi(\theta)$,

$$N_V \int x_i^V(\theta) \psi(\theta) \Pi(v(\theta) | \theta) dG(\theta) = X_i^V \left(\mathbb{E}[\psi(\theta)] + \text{Cov} \left(\frac{x_i^V(\theta)}{\bar{x}_i^V}, \psi(\theta) \right) \right), \quad (18)$$

with the expectation and covariance taken over realized arrivals. Applying (18) with ψ equal to the income-effect term $t_1 \frac{\partial x_1^V}{\partial I} + t_2 \frac{\partial x_2^V}{\partial I}$ and to the extensive-margin term $\bar{\tau} \epsilon_{ext}^V$, we can write this efficiency condition in terms of the elasticities of aggregate demand:

$$\begin{aligned} 0 = & X_i^V + \left(\frac{t_{-i}}{1+t_{-i}} - \frac{t_i}{1+t_i} \right) X_i^V \mathcal{E}_i^V + \frac{t_{-i}}{1+t_{-i}} X_i^V \mathcal{E}_{i,o}^V + \left(\frac{t_{-i}}{1+t_{-i}} - \frac{t_i}{1+t_i} \right) X_i^R \mathcal{E}_{i|z}^R \\ & - X_i^V \mathbb{E} \left[t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right] - X_i^V \text{Cov} \left(\frac{x_i^V(\theta)}{\bar{x}_i^V}, t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right) \\ & - X_i^V \mathbb{E} [\bar{\tau}(\theta) \epsilon_{ext}^V(\theta)] - X_i^V \text{Cov} \left(\frac{x_i^V(\theta)}{\bar{x}_i^V}, \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \right). \end{aligned} \quad (19)$$

Re-arranging equation (19) and setting $\frac{\partial x_i^V(\theta)}{\partial I} = 0$ and $\Pi(v(\theta) | \theta) = 1$, we can easily obtain equations (8) and (9).

Allowing for extensive margin responses (maintaining $\frac{\partial x_i^V(\theta)}{\partial I} = 0$ for now), write $s_i \equiv \frac{t_i}{1+t_i}$ and divide equation (19) by X_i^V :

$$\begin{aligned} 0 = & 1 + (s_{-i} - s_i) \left[\mathcal{E}_i^V + \frac{X_i^R}{X_i^V} \mathcal{E}_{i|z}^R \right] + s_{-i} \mathcal{E}_{i,o}^V \\ & - \mathbb{E} [\bar{\tau}(\theta) \epsilon_{ext}^V(\theta)] - \text{Cov} \left(\frac{x_i^V(\theta)}{\bar{x}_i^V}, \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \right) \end{aligned} \quad (20)$$

Differencing the $i = 1$ and $i = 2$ conditions, the expectation terms cancel, leaving

$$s_1 (\mathcal{E}_{2,o}^V + \mathcal{A}) = s_2 (\mathcal{E}_{1,o}^V + \mathcal{A}) + \text{Cov} \left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \right) \quad (21)$$

where $\mathcal{A} = \mathcal{E}_1^V + \mathcal{E}_2^V + \frac{X_1^R}{X_1^V} \mathcal{E}_{1|z}^R + \frac{X_2^R}{X_2^V} \mathcal{E}_{2|z}^R$ is the attenuation factor. (Dividing equation (21) through by $s_2 (\mathcal{E}_{2,o}^V + \mathcal{A})$ expresses the tax ratio equivalently as the substitution-only ratio of equation (10) plus a single extensive margin component proportional to this covariance.) Now note that $\bar{\tau}(\theta) = s_1 \alpha_1(\theta) + s_2 \alpha_2(\theta)$, with $\alpha_i(\theta)$ the tax-inclusive spending shares defined in section 2.3. By linearity of the covariance, the covariance in

equation (21) splits into an s_1 -weighted and an s_2 -weighted piece; collecting the s_1 and s_2 terms then yields

$$\begin{aligned} s_1 \left[\mathcal{E}_{2,o}^V + \mathcal{A} + \text{Cov} \left(\frac{x_1^V(\theta)}{\bar{x}_1^V} - \frac{x_2^V(\theta)}{\bar{x}_2^V}, \alpha_1(\theta) \epsilon_{ext}^V(\theta) \right) \right] \\ = s_2 \left[\mathcal{E}_{1,o}^V + \mathcal{A} + \text{Cov} \left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \alpha_2(\theta) \epsilon_{ext}^V(\theta) \right) \right] \end{aligned} \quad (22)$$

which is equation (11). Since $\alpha_1(\theta) + \alpha_2(\theta) = 1$, the difference between the two covariance terms is $\text{Cov} \left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \epsilon_{ext}^V(\theta) \right)$, which delivers the sign characterization in the main text, provided both bracketed terms in equation (22) are positive. Allowing for income effects, the same steps append the covariance $\text{Cov} \left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right)$ to the right-hand side of equation (22). Since $t_i \frac{\partial x_i^V(\theta)}{\partial I}$ carries the wedge s_i just as $\alpha_i(\theta) \epsilon_{ext}^V(\theta)$ does, it splits and collects across the two brackets in the same way, replacing each $\alpha_i(\theta) \epsilon_{ext}^V(\theta)$ with $\alpha_i(\theta) \epsilon_{ext}^V(\theta) + (1 + t_i) \frac{\partial x_i^V(\theta)}{\partial I}$. These two additions sum to type θ 's marginal propensity to spend in the destination, so income effects favor a lower rate on the good disproportionately consumed by visitors who spend more of a marginal dollar there.

Finally, consider the aggregate demand interpretation discussed in section 2.3. Differentiating $X_i^V = N_V \int x_i^V(\theta) \Pi(v(\theta) | \theta) dG(\theta)$ and using equation (17),

$$\frac{\partial X_i^V}{\partial t_j} = N_V \int \left[\frac{\partial x_i^V(\theta)}{\partial t_j} \Big|_u - \frac{x_i^V(\theta) x_j^V(\theta)}{(1 + t_1) x_1^V(\theta) + (1 + t_2) x_2^V(\theta)} \epsilon_{ext}^V(\theta) \right] \Pi(v(\theta) | \theta) dG(\theta).$$

The arrival contribution to the aggregate response matrix is symmetric in i and j by inspection; with no visitor income effects, aggregate visitor demand therefore retains the Slutsky symmetry used above. The type-level income effect contribution $-x_j^V(\theta) \frac{\partial x_i^V(\theta)}{\partial I}$ is not generally symmetric in i and j , which is why, once income effects are admitted, the relative rate rule has no symmetric Corlett-Hague representation using only aggregate own-price and leakage elasticities. A representation in terms of the full aggregate Marshallian response matrix remains available. In elasticity form, applying equation (18),

$$\hat{\mathcal{E}}_i^V \equiv -\frac{1 + t_i}{X_i^V} \frac{\partial X_i^V}{\partial t_i} = \mathcal{E}_i^V + \mathbb{E} \left[\frac{x_i^V(\theta)}{\bar{x}_i^V} \alpha_i(\theta) \epsilon_{ext}^V(\theta) \right],$$

and, using the visitor adding-up condition preceding equation (16),

$$\hat{\mathcal{E}}_{i,o}^V \equiv \frac{1}{X_i^V} \left[(1 + t_1) \frac{\partial X_i^V}{\partial t_1} + (1 + t_2) \frac{\partial X_i^V}{\partial t_2} \right] = \mathcal{E}_{i,o}^V - \mathbb{E} \left[\frac{x_i^V(\theta)}{\bar{x}_i^V} \epsilon_{ext}^V(\theta) \right].$$

These two elasticities absorb the arrival response in different amounts. The own-price elasticity absorbs the share $\alpha_i(\theta)$ of good i 's arrival leakage, the part deterred by good i 's own price, whereas the leakage elasticity

absorbs all of it. Add them, and split the full leakage using $\epsilon_{ext}^V(\theta) = [\alpha_1(\theta) + \alpha_2(\theta)] \epsilon_{ext}^V(\theta)$. The α_1 portion cancels against the own-price absorption for good 1, and since $\mathbb{E} \left[\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V} \right] = 0$, the remainder is a covariance:

$$\begin{aligned} \hat{\mathcal{E}}_{1,o}^V + \hat{\mathcal{E}}_1^V + \hat{\mathcal{E}}_2^V &= \mathcal{E}_{1,o}^V + \mathcal{E}_1^V + \mathcal{E}_2^V \\ &\quad + \text{Cov} \left(\frac{x_2^V(\theta)}{\bar{x}_2^V} - \frac{x_1^V(\theta)}{\bar{x}_1^V}, \alpha_2(\theta) \epsilon_{ext}^V(\theta) \right) \\ \hat{\mathcal{E}}_{2,o}^V + \hat{\mathcal{E}}_1^V + \hat{\mathcal{E}}_2^V &= \mathcal{E}_{2,o}^V + \mathcal{E}_1^V + \mathcal{E}_2^V \\ &\quad + \text{Cov} \left(\frac{x_1^V(\theta)}{\bar{x}_1^V} - \frac{x_2^V(\theta)}{\bar{x}_2^V}, \alpha_1(\theta) \epsilon_{ext}^V(\theta) \right) \end{aligned}$$

so equation (11) can be written as equation (10) with $\mathcal{E}_i^V$ replaced by $\hat{\mathcal{E}}_i^V$ and $\mathcal{E}_{i,o}^V$ replaced by $\hat{\mathcal{E}}_{i,o}^V$, the resident terms unchanged. Note, the two representations agree bracket by bracket rather than term by term. What appears as a covariance in equation (11) is the residual left once the arrival responses have been divided between the attenuation factor and the leakage elasticities.

The efficient level follows from the same objects. The combination of efficiency conditions in section 2.4 is the revenue effect of raising both tax-inclusive prices in the same proportion. Resident compensated demands do not move along such a reform: two-good adding-up gives $\sum_i (1 + t_i) \frac{\partial x_j(z)}{\partial t_i} \Big|_{u,z} = 0$, so every resident term drops out. What remains is the mechanical visitor revenue and the row sums of the aggregate visitor response matrix,

$$\sum_{i=1}^2 (1 + t_i) \frac{dR}{dt_i} \Big|_{dT(z)=-x_i(z)dt_i} = \sum_{i=1}^2 (1 + t_i) X_i^V + \sum_{j=1}^2 t_j X_j^V \hat{\mathcal{E}}_{j,o}^V.$$

Setting this to zero and dividing by $t_1 X_1^V + t_2 X_2^V$ gives equation (12). This calculation uses only the row sums of the response matrix and never its symmetry, so it does not require the assumption of no visitor income effects. Read as total responses, the derivatives in $\hat{\mathcal{E}}_{j,o}^V$ already include them. Unbundling the leakage elasticities into their substitution, arrival, and income components gives the explicit form

$$\frac{t_1 X_1^V + t_2 X_2^V}{(1 + t_1) X_1^V + (1 + t_2) X_2^V} = \frac{1 - IE^V - EM^V}{-\frac{t_1 X_1^V}{t_1 X_1^V + t_2 X_2^V} \mathcal{E}_{1,o}^V - \frac{t_2 X_2^V}{t_1 X_1^V + t_2 X_2^V} \mathcal{E}_{2,o}^V} \quad (23)$$

where

$$\begin{aligned} EM^V &= \mathbb{E} [\bar{\tau}(\theta) \epsilon_{ext}^V(\theta)] \\ &\quad + \text{Cov} \left(\frac{(1 + t_1) x_1^V(\theta) + (1 + t_2) x_2^V(\theta)}{\bar{c}^V}, \bar{\tau}(\theta) \epsilon_{ext}^V(\theta) \right) \end{aligned}$$

collects the arrival responses,

$$IE^V = \mathbb{E} \left[t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right] + \text{Cov} \left(\frac{(1+t_1)x_1^V(\theta) + (1+t_2)x_2^V(\theta)}{\bar{c}^V}, t_1 \frac{\partial x_1^V(\theta)}{\partial I} + t_2 \frac{\partial x_2^V(\theta)}{\partial I} \right)$$

collects the income effects, and $\bar{c}^V \equiv [(1+t_1)X_1^V + (1+t_2)X_2^V] / \bar{N}_V$ is mean visitor spending among realized arrivals. Both forms presume visitor tax payments $t_1X_1^V + t_2X_2^V$ are nonzero, and equation (23) additionally requires that the revenue-weighted average of $\mathcal{E}_{1,o}^V$ and $\mathcal{E}_{2,o}^V$ in its denominator not vanish. Where that average vanishes, equation (12)—which bundles substitution, arrival, and income responses into the total leakage elasticities—remains the primitive characterization.

B A Small-Open-Economy Foundation

This appendix shows that the model of the main text is, to first order, the reduced form of a small open economy with production, trade, endogenous producer prices, and an endogenous wage, as described in Section 2.5. Taken at face value, the main-text model violates Walras’s law, as visitor spending in the destination economy does not generate income in the destination. In this section, we introduce a small open economy model where tourist spending finances the purchases of imports by residents, and where producer prices and resident wages are endogenous. I show that, if the compensating reforms of the main text are suitably augmented and if resident demand elasticities are appropriately redefined, the characterization of the optimal tax policy remains unchanged.

In particular, the income tax adjustment must not only compensate residents for the direct cost-of-living effect of a tax change but also replicate the induced wage movement, while the price of the traded good adjusts to hold resident compensated demand for that good fixed. Along these augmented reforms, the revenue ledger is identical to the fixed-price ledger of the main text, and every derivation of the main text and Appendix A applies with the two amendments recorded in Section 2.5: resident elasticities are read as conditional on the traded good, and wedges are read on consumer prices.

Goods 1 and 2 are nontraded. A third good, good 0, is traded at an exogenous world price normalized to one; good 0 is the numéraire. Let $D_i \equiv X_i^R + X_i^V$ denote total demand for nontraded good i and Y_0 domestic output of the traded good. Resident labor is the only primary factor, and production possibilities

are summarized by the labor requirement

$$\Gamma(Y_0, D_1, D_2) = L, \quad (24)$$

with Γ increasing, convex, and homogeneous of degree one; the main text is the special case $\Gamma = a(Y_0 + D_1 + D_2)$. The one new ingredient is that the marginal labor requirements Γ_k vary with the output mix, letting tourist demand move producer prices and the wage with a single factor. I assume throughout that production is diversified—all three goods are produced in strictly positive quantities—and that the equilibrium mapping from statutory taxes to prices and quantities is locally single-valued and differentiable. Perfect competition then requires $p_k = \omega \Gamma_k$ for each good, where ω is the wage. The world market still fixes $p_0 = 1$, but no longer the wage: $\omega = 1/\Gamma_0$ and $p_i = \Gamma_i/\Gamma_0$ move with the output mix. Euler's theorem delivers zero profits,

$$Z \equiv \omega L = Y_0 + p_1 D_1 + p_2 D_2, \quad (25)$$

where Z is aggregate resident labor income. Differentiating equation (25) and subtracting the differentiated production constraint ($dY_0 + p_1 dD_1 + p_2 dD_2 = \omega dL$, by competitive pricing) leaves

$$D_1 dp_1 + D_2 dp_2 = L d\omega = Z d \ln \omega. \quad (26)$$

This identity is the only production-side fact used below: whatever a reform does to prices and the wage, the revaluation of nontraded output equals the change in labor income, exactly.

Consumer prices are $q_0 = 1 + t_0$ and $q_i = p_i + t_i$. A type w resident supplies labor ℓ , earns taxable income $z = \omega \ell$, and solves

$$\max_{\ell, x_0, x_1, x_2} \left\{ u(\phi(x_0, x_1, x_2), \ell; w) : \sum_{k=0}^2 q_k x_k = \omega \ell - T(\omega \ell) \right\},$$

with utility over labor rather than income because the wage now moves. Weak separability extends to good 0, and conditional demands $x_k(z)$ are homogeneous across residents at a given taxable income. Good 0 carries the destination-based use tax t_0 , applying to resident *use* whether a unit is produced domestically or imported. Visitors are exactly as in the main text: they do not purchase good 0 in the destination ($X_0^V = 0$; their purchases of traded goods at home are part of outside consumption x_o), and q_0 enters neither visitor demands nor $v(\theta)$ nor Π . Total tax revenue is $R \equiv N_R \int T(z(w)) dF(w) + \sum_{k=0}^2 t_k X_k^R + \sum_{k=1}^2 t_k X_k^V$.

Because a statutory tax change and a consumer-price change are no longer the same thing, all elasticities in

this appendix are defined with respect to consumer prices, with the main text's leading minus sign:

$$\mathcal{E}_{ik|z}^R \equiv -\frac{q_k}{X_i^R} \frac{\partial X_i^R}{\partial q_k} \Big|_{u,\ell}, \quad \mathcal{E}_{ik}^V \equiv -\frac{q_k}{X_i^V} \frac{\partial X_i^V}{\partial q_k} \Big|_u, \quad s_k \equiv \frac{t_k}{q_k}.$$

With fixed producer prices, $q_k = 1 + t_k$ and these reduce to the corresponding definitions of the main text and Appendix A. The resident conditioning is now on labor and utility rather than taxable income: with the wage endogenous, $z = \omega\ell$ moves with ω , so holding ℓ fixed is not the same as holding z fixed. I keep the subscript z to match the main text, since with producer prices fixed the two coincide. Note, under diversification statutory taxes can implement any local movement in consumer prices— $dt_0 = dq_0$ and $dt_i = dq_i - dp_i$, where $(dp_1, dp_2, d\omega)$ are the equilibrium price responses—so we may treat consumer prices as the reform instruments directly. I assume throughout that these aggregate compensated responses exist and are continuous, that all tax bases are strictly positive, and that the resident own-price response for the traded good is strictly negative, so $\mathcal{E}_{00|z}^R > 0$: negative semidefiniteness alone delivers only a weak inequality, and the companion adjustment and conditional elasticities below divide by $\mathcal{E}_{00|z}^R$.

Aggregating resident budget constraints, using zero profits (equation (25)), and collecting terms yields the economy's current-account identity:

$$q_1 X_1^V + q_2 X_2^V = X_0^R + R - Y_0. \quad (27)$$

Tourist spending on nontraded goods finances residents' use of the traded good and the government surplus, net of domestic traded-good production. This is the accounting the main text was missing: given trade balance, the government budget constraint and the economy's resource constraint are equivalent, and Walras's law is restored. It also confirms that revenue remains the correct objective for the compensated planner: visitors pay producer prices that cover the competitive resource cost of what they buy, so the destination's net gain per unit sold is exactly the tax wedge.

Now the reforms. Fix an arbitrary change (dq_1, dq_2) in home-good consumer prices and pair it with two companion adjustments. First, the price of good 0 adjusts to hold aggregate compensated resident demand for good 0 fixed ($dX_0^R = 0$):

$$\frac{dq_0}{q_0} = -\frac{\mathcal{E}_{01|z}^R \frac{dq_1}{q_1} + \mathcal{E}_{02|z}^R \frac{dq_2}{q_2}}{\mathcal{E}_{00|z}^R}. \quad (28)$$

Second, the income tax compensates residents and replicates the wage:

$$\delta T(z) = -\sum_{k=0}^2 x_k(z) dq_k + [1 - T'(z)] z d \ln \omega, \quad (29)$$

where δT is the change in the schedule at pre-reform incomes. The first piece is the cost-of-living offset of the main text. The second piece is what makes these reforms *replicating* rather than merely compensating: at any fixed labor choice, taxable income moves mechanically with the wage ($dz = z d \ln \omega$), so after-tax income changes by $[1 - T'(z)] z d \ln \omega - \delta T(z) = \sum_k x_k(z) dq_k$ —exactly the cost of the pre-reform bundle at the new prices. The value of the second-stage consumption problem is therefore unchanged at every labor choice, so labor supply and utility are unchanged to first order. Residents are compensated not just for the statutory tax change but for every price and wage movement it induces. This is the one-factor analogue of the price-replicating reforms of Jacquet and Lehmann [2025]: with one primary factor, the income tax alone spans the factor-price direction.

Now the revenue ledger. Income-tax revenue changes by $dR_T = -\sum_k X_k^R dq_k + Z d \ln \omega$ (the schedule change plus incomes moving along the schedule). Adding the mechanical and behavioral changes in commodity revenue, substituting $dt_i = dq_i - dp_i$, and cancelling $D_1 dp_1 + D_2 dp_2$ against $Z d \ln \omega$ by equation (26) leaves

$$dR = t_0 dX_0^R + \sum_{i=1}^2 [X_i^V dq_i + t_i (dX_i^R + dX_i^V)]. \quad (30)$$

This is the fixed-price revenue ledger of the main text, in consumer-price coordinates. Endogenous producer prices and the endogenous wage have not been assumed away: their incidence on residents is offset by the income tax, and their revenue revaluation cancels through competitive-production accounting. The first term vanishes by construction, since the companion adjustment (28) holds the use-tax base fixed. Note, no companion adjustment touches a visitor-side object—visitors buy no good 0, and δT applies to residents—so every visitor-side term is exactly as in the main text. And because residents' utility is unchanged to first order, Pareto efficiency requires that no direction of reform raise revenue. This is the one-factor instance of the central result of Jacquet and Lehmann [2025]: endogenous factor prices do not alter the test for Pareto-improving tax reforms, provided the reforms which neutralize their incidence are available.

It remains to express the resident responses. Along the reform, resident substitution runs through a conditional two-good system: substituting equation (28) into the resident responses gives $dX_i^R = -X_i^R \left[\tilde{\mathcal{E}}_{i1|z}^R \frac{dq_1}{q_1} + \tilde{\mathcal{E}}_{i2|z}^R \frac{dq_2}{q_2} \right]$, where

$$\tilde{\mathcal{E}}_{ik|z}^R \equiv \mathcal{E}_{ik|z}^R - \frac{\mathcal{E}_{i0|z}^R \mathcal{E}_{0k|z}^R}{\mathcal{E}_{00|z}^R} \quad (31)$$

are the conditional elasticities discussed in Section 2.5. In derivative units the conditional system inherits Slutsky symmetry from the full one, and three-good adding-up ($\sum_{k=0}^2 \mathcal{E}_{ik|z}^R = 0$, from homogeneity) implies two-good adding-up: $\tilde{\mathcal{E}}_{i1|z}^R + \tilde{\mathcal{E}}_{i2|z}^R = -\mathcal{E}_{i0|z}^R + \mathcal{E}_{i0|z}^R = 0$. The conditional system thus has exactly the properties—symmetry and adding-up—that resident demand had in the fixed-price model.

Nothing else needs to change. Equation (30) is, term by term, the object differentiated in the main text and Appendix A, with resident substitution given by a compensated two-good system satisfying symmetry and adding-up. Every derivation there goes through unchanged once $\mathcal{E}_{i|z}^R$ is replaced by $\tilde{\mathcal{E}}_{i|z}^R \equiv \tilde{\mathcal{E}}_{ii|z}^R$ and each wedge $\frac{t_i}{1+t_i}$ is read as $s_i = t_i/q_i$, with each tax-inclusive price $1 + t_i$ appearing outside a wedge—in the spending shares $\alpha_i(\theta)$, the extensive margin elasticity $\epsilon_{ext}^V(\theta)$, and mean visitor spending $\bar{c}^V$ —likewise read as q_i : with these substitutions, the efficiency conditions (8)–(9), the differentiation rule (10), and the efficient level (12) follow. The extensions require no separate argument. Extensive-margin responses and visitor income effects are visitor-side objects, and the visitor side of the problem is untouched: visitors do not consume the traded good in the destination, and no companion adjustment enters their prices, their demands, or their arrival decisions. Equations (11) and (23) therefore follow as well. Everything complicated in the fixed-price derivations concerns visitors. The open economy adds one complication on the resident side, and the augmented reforms neutralize it.

Finally, consider the use tax. The reforms above never move q_0 on its own. Raising q_0 alone ($dq_1 = dq_2 = 0$, which requires $dt_i = -dp_i$), with the income-tax companion, completes the system: together the three directions span all local consumer-price reforms, and hence—under diversification—all local commodity-tax reforms. The visitor side of this reform is zero, term by term: visitors purchase no good 0 (no mechanical term), and q_0 is absent from their demands (no substitution term) and from $v(\theta)$ and Π (no participation term). What remains is resident substitution, and Slutsky symmetry converts it to row-0 form: stationarity requires

$$0 = s_0 \mathcal{E}_{00|z}^R + s_1 \mathcal{E}_{01|z}^R + s_2 \mathcal{E}_{02|z}^R, \quad (32)$$

which rearranges to equation (13) of the main text. Note, when good 0 is a net substitute for both home goods, the implied use-tax wedge is a weighted average of the two home-good wedges; and when the home goods carry a common wedge $s_1 = s_2 = s$, row-0 adding-up gives $s_0 = s$: the uniform benchmark of Section 1, a common wedge on *all* resident consumption. This parallels the finding of Jacquet and Lehmann [2025] that Atkinson-Stiglitz uniformity survives endogenous factor prices whenever the neutralizing reforms can be implemented, though they reach commodity taxation by reinterpreting intermediate goods rather than through consumer prices.